

Syllabus of EEE 491 Electrical and Electronics Engineering Design I

Catalog Description: Senior design project involving design and implementation of a complete electrical and electronics engineering system. Development involving multiple areas of electrical and electronics engineering. Simulations. Prototype development and testing. Technical communications and teamwork skills enrichment.

Bilkent Credits: 3, ECTS Credits: 6

Brief Description: A core component of this course is a hands-on design project: the development of a Spoken Number Recognition system implemented on an FPGA platform. This project will provide practical experience in:

- Circuit design and analog circuit simulation
- Signal and speech processing techniques
- MATLAB-based algorithm development
- VHDL coding and digital simulation
- FPGA implementation and debugging

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Lecture Hours / Location:	Wednesday, 10:30 – 11:20, 11:30 – 12:20, room EE-04
Lab Hours / Location:	Saturday, 13:30 – 17:20, room EE-LAB103
Course Materials:	Software: MATLAB, Xilinx Vivado, LT-Spice, a PCB layout editor Hardware: Basys-3 FPGA development board Reading: Properties of audio signals and speech recognition, user manuals of the tools, datasheets of FPGA building blocks, and electronic components. Lecture notes: slides to be uploaded
Prerequisites:	EEE 212 (microprocessors) EEE 313 (electronic circuit design) EEE 321 (signals and systems)
Closely Related:	EEE-102 (introduction to digital circuit design) EEE-211 (analog electronics) EEE-314 (digital electronics) EEE-424 (digital signal processing) EEE-492 (digital signal processing laboratory) ENG-401 (technical report writing)
Workload:	Individual or group work: 12 hours/week (average) Course: 2 hours/week Laboratory: 4 hours/week (team work: three students per group) Project Preparation and Demo: 40 hours Project Report: 20 hours Quiz/Homework: 4 x 1 hours + 4 x 1 hours (preparation)

What you will learn are:

- Conduct lab experiments in preparation for the term project; demonstrate and report on the results obtained.
- Develop mathematical models and conduct analysis for complex engineering problems.
- Design and implement a complete electrical and electronics engineering system involving multi-disciplinary work as a team.

Objectives:

- Make research on “audio signals” and their characteristics, including some details on word/number recognition techniques; like Mel-frequency Cepstrum algorithm.
- Model, simulate, and verify your system design using MATLAB.
- Design a microphone amplifier circuit with filter and simulate by analog simulator.
- Implement the analog circuit by drawing PCB layout, manufacturing, and populating components.
- Digitize audio signals by analog to digital converter (ADC) and capture the digital signals by FPGA.
- Implement the system building blocks/components on FPGA using the hardware design language VHDL.
- Run logic simulations for FPGA. Write test benches in VHDL and use FPGA simulator tool.
- Include design for testability and debugging methods. Implement a debugger in order to transfer data from FPGA to MATLAB.
- Design, implement, simulate, test, integrate, and re-test each building block of your system.
- Complete spoken number recognition system implementation on the Basys-3 FPGA board.
- Apply project management concepts by working in groups during the system integration.
- Describe your design with verification results in project report documents during the system development.

Practices you will have are:

- Apply signal processing, speech processing, MATLAB modeling, and finite-state machine design.
- System development techniques (V-model process, requirement management, verification, and validation).
- FPGA implementation techniques (FPGA design flow, IP core, VHDL coding, Vivado synthesis, simulation, integration, debugging, testability).
- Project management concepts.

Required Materials:

BASYS3: FPGA development board which is composed of a printed circuit board with an FPGA on it. Basys-3 is to be used with the FPGA development tool Vivado Design Suit from AMD (formerly Xilinx) running on a PC. Each group will use one Basys-3 board. The board can be purchased from METEKSAN Bookstore. Most of the lab assignments will be demonstrated on this board.

VIVADO Design Suit (Vivado ML Standard Edition): Design tool that provides high-level design entry with a hardware description language (VHDL), design synthesis, functional and timing simulations, placement, and routing of the logic cells, and configuration code for the FPGA. Download and install the “Vivado ML Standard Edition” software on your personal computer. It is free and you may need to create an account.

LTSpice: Analog simulator software from AnalogDevices. Download and install the software on your personal computer. It is free.

A schematic and PCB Layout Editor: To draw schematic of your electronic board and draw PCB layout. The tool generates files to print the PCB at the laboratory. A free version of an editor can be downloaded found from the internet and choose any of them. Examples are KiCAD, Proteus, DipTrace, CircuitMaker (from Altium), Autodesk EAGLE, Fritzting, DesignSpark PCB, EasyEDA, UpVerter, PCBWeb Designer, ExpressPCB Plus, and TinyCAD.

Lab Work:

Attend the lab hours and study the lab assignments together with your group members. Lab assistants will join the lab hours at any time and they will observe study of each student and grade performance during the semester.

Lab assignments require research, design, simulation, and demonstration. Each team demonstrates lab assignment to the assistant at lab hub at scheduled date.

Lab hub is available 7/24 and you may use the lab whenever it is available.

Lab Assignments:

Lab Assignment	Context
Lab-DEBUG	Design a UART on FPGA in order to transfer a RAM data to PC via USB interface, and analyze the data on PC, typically by using MATLAB.
Lab-CTRL	Data flow controller of the overall system on FPGA. Control the data flow using start signal to each block and status (ready) signal from each block on FPGA.
Lab-ADC	Digitize audio signal by ADC module using the SPI interface. Store the samples in a RAM on FPGA. Transfer RAM data to PC by using the Lab-DEBUG. Analyze the data on MATLAB.
Lab-PCB	Design a PCB for microphone amplifier and anti-aliasing filter circuit. Assemble the PCB and interface with the Lab ADC. Speak to microphone and analyze the data on MATLAB running on the PC.
Lab-WINDOW	Read the sampled data stored in RAM of Lab-ADC by dividing the sampled data in to frames. Multiply each frame by a windowing function in time domain. Store the result in a RAM on FPGA. Use Lab-DEBUG to analyze data on MATLAB.
Lab-FFT	Use FFT IP from Vivado library and design a Fast-Fourier-Transform function on FPGA. Take FFT of the frame data in Lab-WINDOW. Store the result in a RAM on FPGA. Use Lab-DEBUG to analyze data on MATLAB.
Lab-MEL	Design multiple MEL filter-banks in frequency domain on FPGA. Read the FFT data and apply the multiple bandpass filters. Calculate energy at each filter bank output and store energies in a RAM on FPGA. Use Lab-DEBUG to analyze data on MATLAB.
Lab-DCT	Design a Discrete-Cosine-Transform (DCT) function on FPGA. Take DCT of the data from Lab-MEL and store the results, which are the feature vector of a frame, in a RAM on FPGA. Use Lab-DEBUG to analyze data on MATLAB.
Lab-COMPARE	Read the feature vectors from Lab-DCT. Store all the feature vectors in a RAM. Compare the feature vectors with the ones stored before, make a decision for the spoken number, and display it on 7-segment LED on the FPGA board.
Lab-MATLAB	Model the complete term project (spoken number recognition system) and simulate it on MATLAB. Speak to the PC microphone and display the spoken number on PC.

Note that a group can propose a better approach for the recognition algorithm. The proposed alternative solution shall be assessed by the lecturer before implementation and the group shall provide lab assignments (similar to the defined lab assignments) and list specifications of sub-systems.

Grading (tentative):

Task	Weight	Responsible
Quiz/Homework	20%	Student
In-Class Attendance (Lab and Class)	10%	Student
Lab Work (performance)	15%	Student
Term Project (Lab Demonstrations)	35%	Group
Paper(s)/Reports (incl. project demonstrations)	20%	Group

Attendance: For any missed lecture or lab, you lose one point.

Lab Grading Factors per Demonstration:

Lab Assignment	Difficulty Factor
Lab-CTRL	2
Lab-DCT	2
Lab-WINDOW	2
Lab-DEBUG	3
Lab-MATLAB	3
Lab-MEL	3
Lab-ADC	4
Lab-COMPARE	5
Lab-FFT	5
Lab-PCB	5

Late Demonstration Factors:

On-time:	x1.0
1-week-late:	x0.9
2-weeks-late:	x0.7
3-weeks-late:	x0.5
4-weeks-late:	x0.4
more:	x0.0

Degrading Factors:

No Simulation:	x0.0
Circuit on breadboard:	x0.2
Lab-MATLAB performance:	x (number-recognition-rate)
Group member performance:	x (group-member grading and assistant grading)

Sub-Systems used in Final Demonstration	Points Over 100	Possible Deviation
Lab-PCB	20	On Breadboard: 5
Lab-ADC	10	-
Lab-WINDOW	5	-
Lab-FFT	20	On MATLAB: 5
Lab-MEL	15	On MATLAB: 5
Lab-DCT	10	On MATLAB: 5
Lab-COMPARE	10	On MATLAB: 5
Recognition Performance:	10	-

Note that in order to use Lab-MATLAB functions, your Lab-MATLAB assignment must have been demonstrated and checked by assistant. Therefore, if you use your Lab-MATLAB for the project demo then the best you get is 25 points.

Lab Schedule (tentative):

Because lab due dates can fall on Wednesdays, the deadlines listed below may be advanced by three days.

Lab Assignment Study and Lab Demonstration	Due Date	Week Count
No Lab	Sept 19	1
Lab Study	Sept 26	2
Lab-DEBUG	Oct 03	3
Lab-CTRL	Oct 10	4
Lab-ADC	Oct 17	5
Lab-WINDOW	Oct 24	6
No Lab	Oct 31	7
Lab-PCB	Nov 07	8
Lab-MATLAB	Nov 14	9
Lab-FFT	Nov 21	10
Lab Study	Nov 28	11
Lab-MEL	Dec 05	12
Lab-DCT	Dec 12	13
Lab-COMPARE	Dec 19	14
Overall System Final Demo	Jan 11	15

Academic Calendar:

Start of Classes	Sep 16 (Wed)
No Class	Oct 28, 29, 30 (Wed, Thu, Fri)
Last Day of Classes	Dec 25 (Fri)
Final Exams	Dec 26 – Jan 10 (Sunday)
EEE491 Final Demo	Jan 11 (Mon)
Grades announcement	Jan 14 (Thu)

Generative AI (GenAI) Usage Policy:

- VHDL Design and Testbench Code:** Students are permitted to use GenAI tools to generate VHDL design and testbench code for assignments.
- Professional Role:** Each group is required to prepare a comprehensive report detailing their use of GenAI throughout the design process. Students shall act as ****Lead Engineers**** overseeing the AI, maintaining full technical authority and critical oversight rather than acting as passive users.
- Prompt and Chat Documentation:** The report must include the initial problem definitions provided to the GenAI, as well as every subsequent prompt, data input, and clarification. The full, unedited chat history must be documented, including ****Correction Logs**** where the AI's output was audited, disputed, or refined.
- Code Outputs and Citations:** The report must include all raw code outputs generated by the GenAI and a complete list of all tools and models used (e.g., GPT-4o, Claude 3.5, and Gemini 1.5 Pro).
- Submission:** Each group must submit the specified report content by the announced deadlines.
- In-Class Presentation:** Groups will present their reports in person. The presentation must be divided into sections, with each team member presenting a specific portion of the design and AI interaction.
- Collective Responsibility:** All team members are held collectively responsible for the technical accuracy of the report and the actual functionality/synthesizability of the generated code.

8. **Exam Assessment:** To ensure individual mastery, questions specifically related to a group's unique report and their specific GenAI implementation will be included in exams for all members of that group.

Lecture Schedule (tentative):

Date	Topic
Sept 16	Introduction to EEE491, discussion on the project and requirements. Speech Recognition. Introduction to FPGA, FPGA Design Flow.
Sept 23	VHDL practice. FPGA Design Implementation.
Sept 30	Finite-State Machines (FSM), FSM state-transition diagrams. VHDL practice, variable vs. signal assignments.
Oct 7	Circuit design, Amplifier circuits, Operational Amplifier. Low-pass, high-pass, band-pass filters.
Oct 14	PCB layout, R, L, C coupling.
Oct 21	Quantization, Sampling, Aliasing, A/D conversion, ADC datasheet, SPI interface.
Oct 28	No Class (Republic Day Eve)
Nov 04	FFT IP block configuration
Nov 11	Linear Time-Invariant Systems, Fast Fourier Transform, Discrete Time Fourier Transform, Discrete Fourier Transform, Spreading of energy (leakage) and time-domain windowing.
Nov 18	Product Development, System Engineering
Nov 25	Process V-model.
Dec 02	System Modeling.
Dec 09	Requirements Engineering, Testing (Verification and Validation).
Dec 16	Project Management.
Dec 23	System on Chip by FPGA, Embedded systems, FPGA SoC architecture examples, FPGA technology.